KEVIN REN

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Princeton PhD working on high-dimensional probability, geometric analysis, and algorithms, with applications to modern machine learning and statistics.

Education

Princeton University, Princeton, NJ

- Ph.D. in Mathematics (Advisor: Assaf Naor) **Expected May 2027**
- M.S. in Mathematics **Sep 2023**

Massachusetts Institute of Technology, Cambridge, MA

- B.S. in Mathematics and Physics (GPA 5.0/5.0) **May 2022**

Research & Industry Experience

Graduate Researcher, Princeton University **2022–Present**

- Research in geometry, high-dimensional probability, and algorithms with connections to ML theory.
- 17 papers/preprints, including STOC 2025 and SODA 2026; work featured twice in *Quanta Magazine*.
- Developed probabilistic and geometric tools for embeddings, growth bounds, stability, and structure recovery.

Quantitative Research Intern, Jane Street Capital **Summer 2024**

- Analyzed the market impact of high-volatility events (like company earnings reports).
- Designed and implemented a rigorous Python pipeline to evaluate predictive signals on financial data.
- Rapidly iterated on signal-finding methods using principled Bayesian reasoning and hypothesis testing.

Selected Projects

Randomized Clustering for Structured Data

- Proposed a new multi-scale randomized clustering algorithm for datasets with nice geometric structure, achieving improved theoretical guarantees over long-standing baselines.
- Presented at 2026 Symposium on Discrete Algorithms (SODA).

Manifold Reconstruction from Noisy Data

- Designed a new geometry-aware, statistical algorithm to identify low-dimensional manifold structure from noisy data; relevant to generative modeling and denoising.
- Presented at 2025 Harvard Conference on Geometry and Statistics.

Honors & Awards

Simons Foundation Dissertation Fellowship for Mathematics	**2025–27**
Cubist / Point72 PhD Fellowship	**2025–26**
NSF Graduate Research Fellowship (GRFP)	**2022–27**
Alibaba Global Math Competition, Global Top 70	**2021–23**
Putnam Top 25	**2021**
USA Math Olympiad Winner (Akamai Award)	**2015, 2016**

Skills

Programming: Python (numpy/pandas), PyTorch, C++, MATLAB, Mathematica

Languages: English (Native), Chinese (Fluent)